GIVING WINGS TO INDIAN INNOVATION

If the country has to own a piece of the future, it must both deepen research and bring things up to speed in a dynamic tech era

By AMARNATH K. MENON

As the world moves towards a fifth industrial revolution, or IR 5.0, technology has become the launchpad for soaring national ambitions. But growth hits exponential speeds here—in volume, depth and complexity. Is India ready? Not quite. The future is Artificial Intelligence. Combined with the boundless connectivity of 6G, AI will enable a seamless fusion of the digital, physical and human world. Not to talk of Web 3.0, blockchains, the Internet of Things, quantum computing and synthetic biology. In space, a whole vista of aerospace systems is opening up—cis-lunar space (Earth-Moon corridor) as an economic activity; a hybrid of robotics- and human-based space exploration; space tourism beyond the Near-Earth Orbit; drones, for transformative last-minute delivery, and for combat. While India has the awareness and the interest, it is way behind in scientific research, and fostering technologies has remained a work in progress in the absence of appropriate S&T policies and research priorities, besides the lack of funding. As the country takes its place among spacefaring nations, these gaps need to be rapidly filled. ■



EXPERTSPEAK



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"India needs to concentrate on technology adequacy soon in a few critical sectors besides a long-term goal for technology security in strategic sectors"